

23 D consider change in magnetic flux:

$$\begin{aligned}\Delta(BA) &= A(B_{\text{final}} - B_{\text{initial}}) \\ &= -2AB_{\text{initial}} \\ &= -2(25 \times 10^{-4})(2.0 \times 10^{-4}) \sin 60^\circ \\ &= -\left(\frac{\sqrt{3}}{2}\right)(10^{-6})\end{aligned}$$

by Faraday's law, induced e.m.f. E is:

$$\begin{aligned}|E| &= \frac{d}{dt}(N\phi) \\ &= IR = \left(\frac{d}{dt}Q\right)R\end{aligned}$$

$$\Delta N\phi = R\langle Q \rangle$$

$$\begin{aligned}\langle Q \rangle &= \frac{N\Delta\phi}{R} = \frac{N\Delta BA}{R} \\ &= \frac{(500)\frac{\sqrt{3}}{2}(10^{-6})}{5.0} \\ &= 8.66 \times 10^{-5} \text{ C}\end{aligned}$$

24 A 1 period is 5 ms